EVOLUTION

OF ADVANCED MICROPROCESSORS

Name: Aaron Manoj | Class: S2 CSE-A | Roll No.: 1

Timeline

INTRODUCTION

Intel 80386

The first 32-bit processor, enabling modern multitasking and OS support.

1985

Pentium

Introduced superscalar architecture, allowing the CPU to execute two instructions at once.

1993

1982

Intel 80286

Introduced "Protected Mode," allowing for larger memory addressing.

1989

Intel 80486

Integrated the math coprocessor and cache onto a single chip for massive speed gains.

KEY FEATURES

1

Intel 80286

- 16-bit Architecture: Handled data in 16-bit chunks.
- Protected Mode: Allowed access to up to 16 MB of RAM.
- Instruction Efficiency: Performed much faster clock-for-clock than the original 8086.

2

Intel 80386

- 32-bit Data Bus: Doubled the data path width of its predecessors.
- Virtual 8086 Mode: Allowed multiple MS-DOS sessions to run simultaneously.
- Paging Support: Laid the groundwork for modern virtual memory management.

3

Intel 80486

- Built-in FPU: The Floating Point Unit (math coprocessor) was integrated into the CPU.
- L1 Cache: Included 8 KB of on-chip memory to reduce data bottlenecks.
- Burst Mode: Faster data transfer between the CPU and memory.

4

Pentium

- Superscalar Execution: Two execution pipelines (U and V) for parallel processing.
- 64-bit External Data Bus: Doubled the speed of data moving in and out of the CPU.
- Branch Prediction: Used logic to guess which path a program would take, reducing delays.

REAL WORLD IMPACT

1

Intel 80826

Powered the IBM PC/AT, making business accounting and spreadsheets mainstream.

2

Intel 80836

Enabled the success of Windows 3.1, turning the PC into a graphical, multitasking workstation.

3

Intel 80846

Made High-End Desktop Publishing and early 3D gaming (like DOOM) possible.

4

Pentium

Ushered in the Multimedia Era, allowing for smooth video playback and high-quality digital audio.

MAJOR IMPROVEMENTS

Intel 80826	Intel 80836	Intel 80846	Pentium
Speed: 6–12.5 MHz **Word Size**: 16-bit **Cache**: None **Architecture**: 16-bit CISC	**Speed**: 16–33 MHz **Word Size**: 32-bit **Cache**: None **Architecture**: 32-bit CISC	**Speed**: 25–100 MHz **Word Size**: 32-bit **Cache**: 8 KB L1 **Architecture**: Pipelined CISC	**Speed**: 60–233 MHz **Word Size**: 32-bit **Cache**: 16 KB L1 **Architecture**: Superscalar CISC

COMPARISON

Features	Intel 80826	Intel 80836	Intel 80846	Pentium
Year	1982	1985	1989	1993
Word Size	16-bit	32-bit	32-bit	32-bit (Internal)
Speed	6-12 MHz	16-33 MHz	25-100 MHz	60-200+ MHz
Cache	None	External only	8 KB L1	16 KB L1
Special Feature	Protected Mode	Virtual 8086 Mode	Integrated FPU	Superscalar Design

CONCLUSION

The evolution from the 80286 to the Pentium represents the transition of the computer from a specialized office tool to a multipurpose multimedia machine. By moving from 16-bit to 32-bit architecture and integrating cache and math coprocessors directly onto the silicon, Intel paved the way for modern operating systems and the high-speed computing we take for granted today.